

LIBBY CASE REPORT

osteosarcoma-mets-dll3-h7r2

Multi-agent decision support — clinician track

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Libby is an experimental decision-support tool. The recommendations in this report have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.

Executive summary

Case:osteosarcoma-mets-dll3-h7r2**Question:**Metastatic osteosarcoma with user-reported DLL3 RNA expression. What interventions could target DLL3 expression, gated on IHC confirmation? **Evidence base:**7 trials, 5 clinical-evidence rows, 4 preclinical rows. Two ranked rows: a workup gate plus one biomarker-conditional therapeutic option. Board agreement: full consensus on the workup; one persistent dissent (critic) on the conditional rank-2 trial.

What this report covers

The PI's synthesis frames the case around the DLL3 IHC test that gates whether tarlatamab via NCT06788938 is on the table. Standard 2L+ care for relapsed osteosarcoma (regorafenib, cabozantinib) was reviewed by the board but lies outside this case's targetable-feature scope; it is not ranked here.

Top-line findings

- DLL3 IHC SP347 is the load-bearing test. RNA expression alone does not establish membrane DLL3, and every DLL3-directed therapy in clinical use requires IHC. All five personas converged on this without dissent.
- The conditional rank-2 option is tarlatamab via NCT06788938 — a basket trial that includes osteosarcoma. The conservative's earlier toxicity veto lifts on biomarker confirmation.
- The critic's dissent on the trial persists: there are no published osteosarcoma data with tarlatamab, and cross-tumor translation from SCLC is unproven. That's the scientific question this enrollment would actually answer.
- **If IHC is negative:**rank 2 is foreclosed and the case has no within-scope recommendations. Standard 2L+ care for relapsed osteosarcoma exists but lies outside Libby's targetable-feature ranking — that's a separate conversation with the treating team.

Recommendation summary

Shared first step:DLL3 IHC (SP347) on tumor — gates whether rank 2 is reachable. All 5 personas endorse.

1. **Tarlatamab via NCT06788938**— *Conditional on DLL3 IHC positive — foreclosed if test is negative.*Strong preference fit (high-risk-high-reward + prefers trials); critic dissent on cross-tumor translatability persists. The only therapeutic option within scope of this case's targetable feature.

What this report does **not** cover

Standard 2L+ care for relapsed osteosarcoma (regorafenib, cabozantinib) — reviewed by the board but out of scope. Dose adjustments, monitoring schedules, sequencing across lines, payer/access logistics, or institution-specific trial slot availability are also outside scope.

How to use this report

The ranked option carries the board's agreement state (endorsed, dissent, or veto by persona) and 1–3 anchor citations. The biomarker-conditional rank-2 option is flagged inline; the cross-cutting caveat at the top of the case page maps what's foreclosed if the test is negative. If IHC is negative, the next conversation with the treating team is about standard 2L+ care for the indication, separately from Libby.

Libby is an experimental decision-support tool. The recommendations on this page have not been reviewed by a clinician treating this patient. Do not act on this report without consulting a qualified oncologist.

<!-- libby:downloads:begin -->

Downloads

- [Clinician PDF report](#)— ranked recommendations + evidence + sources
- [Patient/caregiver PDF](#)— plain-language summary
- [Master manuscripts table \(PDF\)](#)— every paper considered — n, effect, variance, toxicities
- [Self-contained HTML](#)— recommendations table that opens offline

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Research question

In metastatic osteosarcoma after first-line MAP, what interventions could target DLL3 expression, gated on IHC confirmation?

Patient profile (scrubbed)

- **Primary site / histology:**bone — osteosarcoma
- **Stage:**IV (metastatic)
- **Performance status (assumed):**ECOG 1
- **Age band (assumed):**18-29 (typical osteosarcoma demographics; not user-supplied)
- **Sex:**unknown
- **Biomarkers:****DLL3 — RNA only (confirmation_status: rna_only); IHC SP347 status unknown.**
Decision-relevant resolution: $\geq 1\%$ (preferably $\geq 25\%$) by IHC for DLL3-directed clinical trials.
- **Prior therapy (assumed):**MAP frontline; response not provided.

Preferences

- **Efficacy/toxicity weight:**0.85 (strong efficacy lean)
- **Toxicity vetoes:**none stated
- **Modality constraints:**none stated
- **Free text:**"accepts high-risk high-reward options"
- **Trials preferred:**yes

Scope summary

7 trials, 5 clinical-evidence rows, 4 preclinical rows. Two ranked rows: a workup gate at rank 1 plus one biomarker-conditional therapeutic option at rank 2. Board agreement: full consensus (all five personas) on the workup; one persistent dissent (critic) on the conditional rank-2 trial. Standard 2L+ care for relapsed osteosarcoma (regorafenib, cabozantinib) lies outside this case's targetable-feature scope and is not enumerated here.

Cross-cutting caveat (read first)

The DLL3 RNA expression in user input does not establish membrane DLL3 protein. The DLL3 IHC SP347 test (rank 1) gates whether tarlatamab via NCT06788938 (rank 2) is on the table at all. Every DLL3-directed therapy in current clinical use requires IHC protein-level confirmation; RNA expression is necessary but not sufficient.

- **Rank 2 (tarlatamab)** is conditional on DLL3 IHC $\geq 1\%$ (preferably $\geq 25\%$). It is the only therapeutic option within scope of this case's targetable feature.
- **If IHC is negative:** rank 2 is foreclosed and this case has no within-scope recommendations. The targetable-feature pathway is closed; the patient and clinician should pursue standard 2L+ care for relapsed osteosarcoma through their normal care channel — that lies outside Libby's targetable-feature-driven scope here.
- **Workup logistics:** SP347 IHC is non-toxic, runs on archival tissue (no fresh biopsy required), takes 1–3 weeks, and costs almost nothing relative to a treatment cycle. Confirm assay availability at the treating institution.

Intervention grouping

- **DLL3-directed BiTEs (biomarker-conditional):** tarlatamab via NCT06788938. Cross-tumor efficacy anchor: DeLLphi-301 ([PMID 37861218](#)) and DeLLphi-304 ([PMID 40454646](#)).

Top interventions

Rank 1. DLL3 IHC (SP347) on tumor — diagnostic gate

Non-toxic. Resolves whether the DLL3-directed pathway is open. Required for trial NCT06788938.

Evidence base

NCT06788938 enforces DLL3 IHC $\geq 25\%$ (stage 1) or $\geq 1\%$ (stage 2) for enrollment. The basket-trial design exists precisely because DLL3 RNA expression is not a sufficient enrollment biomarker. Mechanistic anchor: every DLL3-directed therapy in clinical development requires protein-level confirmation.

Likelihood of desired effect

The test resolves whether rank 2 is reachable. Non-toxic and cheap regardless of result.

Toxicity profile

- None. Lab test on tissue.

Counter-productive mechanisms / dissent

Board endorsement was unanimous. No persona dissented or vetoed.

Practical considerations

Archival or fresh tumor; SP347 antibody assay availability at the treating institution should be confirmed. 1–3 week turnaround. The IHC is the precondition for every DLL3-directed action regardless of which therapy the patient ultimately pursues.

Why this rank

The IHC is the precondition for rank 2. The board treated it as the gate, not as a therapy.

Per-trial detail

Test / trial	Efficacy context	Toxicity	Reference
DLL3 IHC SP347 (assay)	Gates enrollment in DLL3-directed therapy trials	None — diagnostic	NCT06788938

Rank 2. Tarlatamab via NCT06788938

Conditional on `dll3_ihc:positive`. Foreclosed if test is negative.

High-efficacy bispecific T-cell engager; preferences point exactly here when the trial is reachable; one persistent dissent on cross-tumor translation.

Evidence base

NCT06788938 (single-arm phase 2 basket, Simon two-stage, n=29 planned) is the trial enrolling osteosarcoma at biomarker-positive resolution. The mechanistic case rests on cross-tumor evidence: DeLLphi-301 (PMID 37861218 ; ORR 40%, n=220 SCLC) and the confirmatory DeLLphi-304 phase 3 (PMID 40454646; OS HR 0.60, 95% CI 0.47–0.77, median OS 13.6 vs 8.3 mo). No published osteosarcoma data with tarlatamab exist.

Likelihood of desired effect

Assuming positive IHC, mechanism-fit is concordant. Whether SCLC's clinical effect transfers to osteosarcoma is the central scientific question this trial enrollment would answer. The user's preferences (efficacy lean 0.85, accepts high-risk-high-reward, prefers trials) point exactly here. **A negative IHC forecloses this rec entirely.**

Toxicity profile

- CRS in ~50% (mostly grade 1–2; grade ≥3 ~1% in SCLC)
- ICANS-like neurologic events ~10%
- Inpatient cycle-1 step-up dosing required for CRS mitigation
- Step-up dosing: 1 mg D1, 10 mg D8/D15, then 10 mg q2w (28-day cycles)

User has no toxicity vetoes; CRS and inpatient cycle-1 chair time are not flagged.

Counter-productive mechanisms / dissent

The critic's dissent persists. No published osteosarcoma data with tarlatamab; cross-tumor translation from SCLC is unproven; the trial's basket design exists to ~~test~~this premise. The conservative's earlier toxicity veto (issued under the IHC-unconfirmed scenario) lifts on biomarker confirmation — its own rationale specified the veto was contingent on IHC. Consensusite's qualified-on-guideline-fit position upgrades to endorsement when the trial-enrollment principle is in play.

Practical considerations

- Trial open at NCT06788938 (recruiting). Slot availability at treating site should be confirmed.
- Trial enrollment is NCCN cat-1 for relapsed osteosarcoma irrespective of mechanism.

- Inpatient cycle-1 monitoring required.
- Off-guideline for osteosarcoma per indication; the trial provides the regulatory pathway.

Why this rank

The only DLL3-directed therapeutic option on the table, conditional on biomarker confirmation. Foreclosed entirely if IHC negative.

Per-trial detail

Therapeutic agent	Efficacy	Toxicity	Reference
Tarlatamab — NCT06788938 DLL3-IHC-selected basket (osteosarcoma included), n=29 planned	ORR endpoint at 18 mo (primary); no osteosarcoma data yet	Per SCLC mechanism: CRS ~50%, ICANS ~10%, inpatient C1	NCT06788938
Tarlatamab — DeLLphi-301 SCLC (cross-tumor mechanism evidence), n=220	ORR 40% (95% CI 29–52); mPFS 4.9 mo	CRS ~50% (G3+ ~1%); ICANS ~10%; G3+ TRAEs 30%	PMID 37861218
Tarlatamab — DeLLphi-304 SCLC (cross-tumor confirmatory), n=509	OS HR 0.60 (95% CI 0.47–0.77), p<0.001; median OS 13.6 vs 8.3 mo	G3+ TRAEs 24% vs 53% chemo arm — favorable vs comparator	PMID 40454646

Classes examined but not ranked

- **Anti-PD-1 / PD-L1 monotherapy in osteosarcoma:**historically negative (SARC028 and successors); not DLL3-directed.
- **HER2-, GD2-, B7-H3-directed therapies:**not in the dossier for this case; no biomarker selection performed at intake.
- **MTP-PE (mifamurtide):**historical adjuvant data in newly diagnosed disease; not 2L+; not DLL3-directed.
- **Multi-kinase TKIs (regorafenib, cabozantinib):**standard 2L+ care for relapsed osteosarcoma per NCCN ([PMID 31013172](#), [PMID 30477937](#), [PMID 32078813](#)). They do not target DLL3 and lie outside this case's targetable-feature scope. Surfaced here for transparency only — they should be considered through the patient's normal care channel, not via Libby's targeted-feature ranking.

Ranked prioritization

```
<div class="trial-table-wrap"> <div class="trial-scroll"> <table class="trial-table">
<thead><tr><th>Rank</th><th>Intervention</th><th>Likelihood of effect</th><th>Toxicity
burden</th><th>Counter-productive MoA</th><th>Overall</th></tr></thead> <tbody> <tr> <td>1</td>
<td><strong>DLL3 IHC (SP347) on tumor — diagnostic gate</strong><br><small><em>endorse:</em> <span
class="persona persona-risktaker">risktaker</span> <span class="persona
persona-conservative">conservative</span> <span class="persona persona-critic">critic</span> <span
class="persona persona-concensusite">concensusite</span> <span class="persona
persona-advocate">advocate</span></small></td> <td>Diagnostic certainty — resolves whether the
DLL3-directed pathway is reachable; required by NCT06788938 (≥25% stage 1, ≥1% stage 2).</td> <td>Low
(none — diagnostic test on archival tissue)</td> <td><strong>N/A</strong> <span class="cpm-desc">(diagnostic,
not therapeutic)</span></td> <td><strong>Non-toxic precondition that gates rank 2 entirely; run regardless of
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which therapy is ultimately chosen.

tarlatamab via NCT06788938 (UCCC-01 basket) (conditional on dll3_ihc positive) endorse: risktaker advocate conservative consensusite dissent: critic Cross-tumor extrapolation: SCLC OS HR 0.60 (DeLLphi-304); ORR ~40% (DeLLphi-301). Osteosarcoma efficacy is the open question NCT06788938 will answer. Moderate (CRS ~50% mostly G1-2; CRS G≥3 ~1%; ICANS-like ~10%; inpatient cycle-1 step-up dosing required) Moderate (On-mechanism CNS bystander T-cell activation drives ICANS; possible DLL3 antigen-loss escape on repeated dosing) The only DLL3-directed option when IHC is positive — preference-aligned but cross-tumor translation untested; foreclosed if IHC negative.	2
regorafenib endorse: risktaker conservative critic consensusite advocate Moderate. mPFS 3.6 vs 1.7 mo (SARC024, HR 0.42, p=0.017); independently replicated in REGOBONE — strongest 2L+ osteosarcoma RCT evidence base. High (G3+ AEs 70-76%: hand-foot, hypertension, fatigue, mucositis; ~65% require dose reduction; no treatment-related deaths) Low (Anti-angiogenic mechanism can impair wound healing and raise bleeding risk; no on-target counter-productive vector) Strongest replicated 2L+ osteosarcoma RCT evidence; biomarker-independent backbone regardless of IHC.	4
cabozantinib endorse: risktaker conservative critic consensusite advocate Modest. ORR ~12% in CABONE osteosarcoma cohort (n=42); mPFS 6.2 mo. Single-arm phase-2 evidence — supportive but not RCT-grade. High (G3+ AEs ~68%: hypertension, hand-foot, hypophosphatemia, transaminitis; thromboembolism ~7%) Low (VEGFR-driven thromboembolism / wound-healing impairment is class-effect off-target; no on-mechanism counter-productive vector) Single-arm sarcoma-cohort signal; biomarker-independent fallback when an RCT-grade option is exhausted or unavailable.	

!!! note "Reading the columns" **Toxicity burden** is patient-level G3+ AE severity (Low / Moderate / High) summarized from the trial publications. **Counter-productive MoA** is the mechanism-level risk that the intervention's own pathway could blunt the therapeutic goal — distinct from patient AEs. The board's endorse / dissent / veto state appears as pills under each intervention; full per-persona rationale lives on the [board page](#).

Caveats

- **Evidence base is small for the conditional trial.** NCT06788938 plans n=29 with no published efficacy data yet. The mechanistic basis is the cross-tumor SCLC evidence (DeLLphi-301 / 304), which is robust in SCLC but unproven in any other tumor type.
- **Biomarker dependency:** rank 2's eligibility assumes a binary IHC result. Indeterminate / weak-positive (1–24%) edge cases are not explicitly addressed; in practice, NCT06788938's stage-2 ≥1% threshold may permit enrollment.

- **What would change the ranking:**
 - A positive DLL3 IHC plus a head-to-head osteosarcoma cohort within the basket trial reading out would move tarlatamab from "mechanism unproven cross-tumor" to "evidence-supported" and tighten its rank-2 confidence.
 - A user toxicity veto on CRS / inpatient cycle-1 monitoring would foreclose tarlatamab even with positive IHC.
 - Slot unavailability at NCT06788938 sites would close the within-scope therapeutic pathway entirely; standard 2L+ care for the indication would be the patient's residual option (outside Libby's enumeration here).
- **Out-of-scope caveat:** standard 2L+ care for relapsed osteosarcoma (regorafenib's two-RCT base, cabozantinib's CABONE phase-2) was surfaced by the trial screener and reviewed by the board, but those drugs do not target the user's stated targetable feature (DLL3). They are not ranked on this page; pursue them through the treating team independent of Libby's targetable-feature ranking.

Sources

PubMed (PMID):

- [37861218](#)— Ahn et al., DeLLphi-301, *NEJM*2023
- [40454646](#)— Mountzios et al., DeLLphi-304, *NEJM*2025
- [35983951](#)— DLL3 IHC reference (cited in workup row)

ClinicalTrials.gov (NCT):

- [NCT06788938](#)— DLL3-IHC-selected basket including osteosarcoma (tarlatamab)
- [NCT05060016](#)— DeLLphi-301 (cross-tumor mechanism)
- [NCT05740566](#)— DeLLphi-304 (cross-tumor confirmatory)

Transparency artifacts

- [Trial table](#)— 7 rows, all 25 columns
- [Evidence list](#)— 5 clinical-evidence rows + 4 preclinical rows
- [Tumor-board transcript](#)— 5 positions, 20 cross-critiques
- [Recommendations table](#)— full ranked detail with biomarker-conditional flag
- [Plain-language summary](#)— patient/caregiver track

Run log

Authored May 2026. Inputs supplied: targetable feature ("DLL3 RNA expression"), clinical descriptor ("metastatic osteosarcoma"), preference summary ("accepts high-risk high-reward options"). Re-rendered when Libby's biomarker-confirmation contract narrowed: instead of emitting parallel positive/negative therapeutic rankings, the PI now flags the confirmatory test as the rank-1 workup and tags only biomarker-conditional recs with :positive. Negative-branch ranking is foreclosed via the cross-cutting caveat above. Standard 2L+ care for the indication is out of scope and not ranked here.

Sources

PubMed (PMID):

- [35983951](#)
- [37861218](#)
- [40454646](#)
- [31013172](#)
- [30477937](#)
- [32078813](#)

ClinicalTrials.gov (NCT):

- [NCT06788938](#)